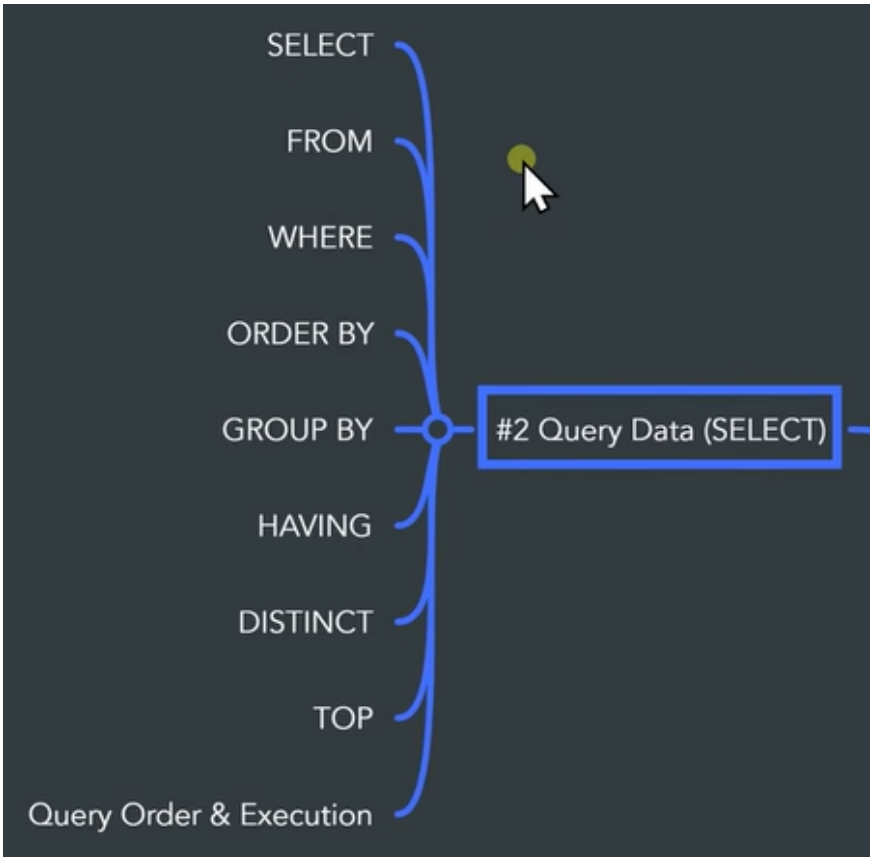
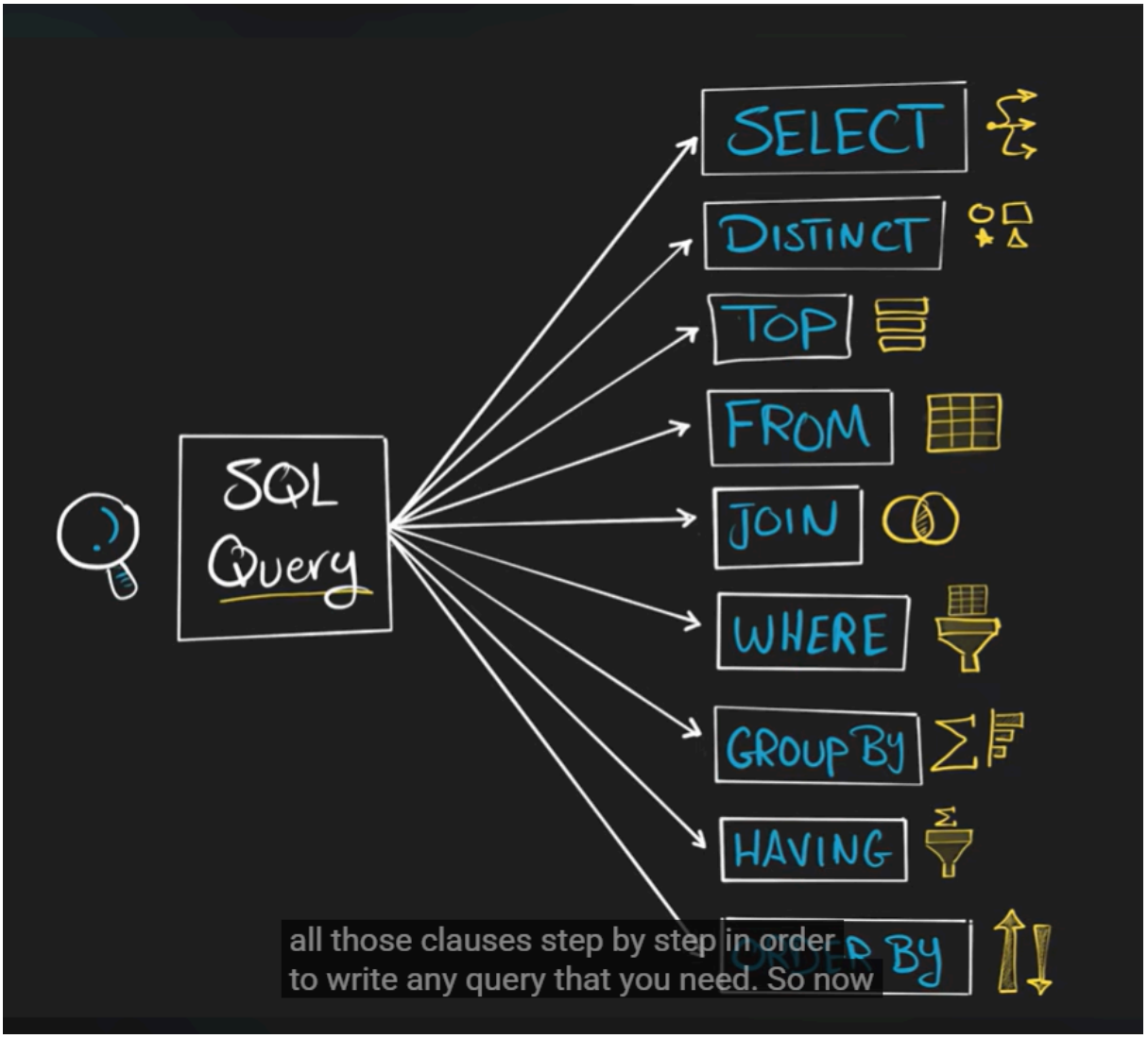
Query Data



Perfect Order



This is the order of using these keywords to write the query qhen all the above are useful

1. Select,From

1. Keyword used to say from which table we need to select the data
2. From tells the Table name
3. Always remember firstly “FROM” keyWord gets executed
4. Select is Operated on Columns of the Table
5. Used to Filter columns
6. “*” → selects all columns ,
7. Select * from table ⇒ retrieves all columns

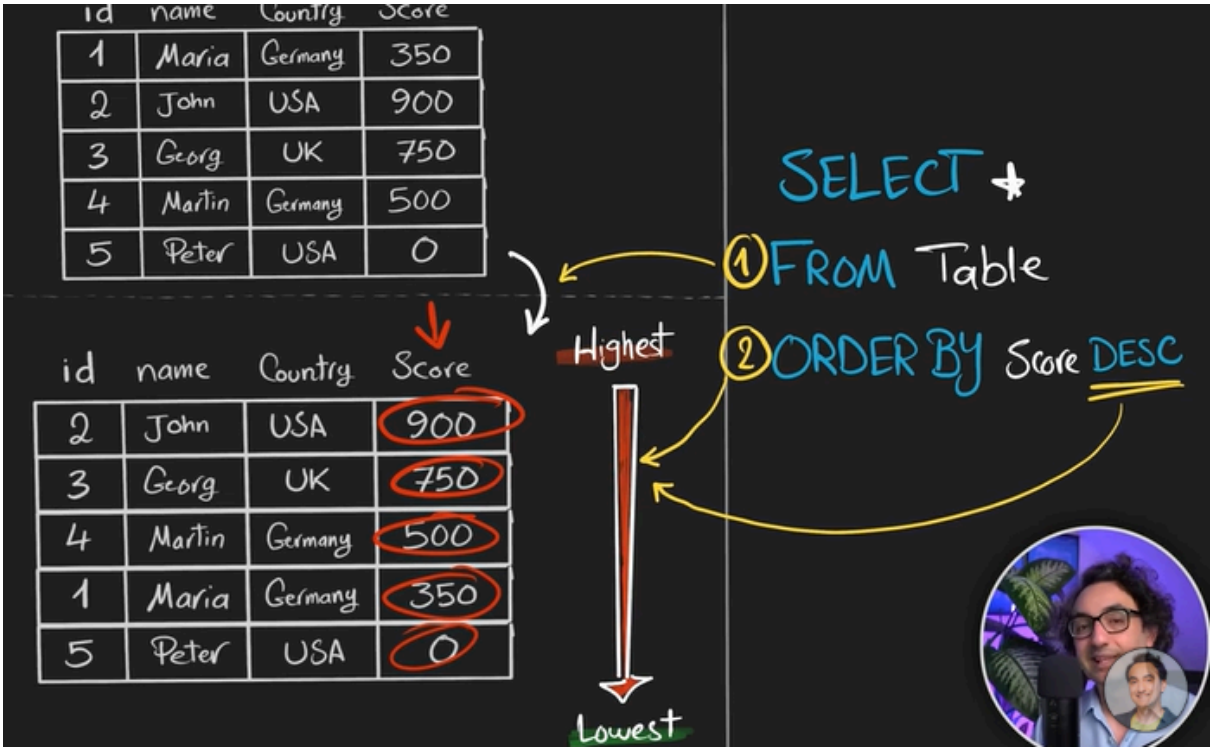
8. Select col1,col2 from table ⇒selects/ retrievs only col1,col2 from table
9. In all cases “SELECT” is executed at Last

2. Where

1. Used to Filter Data based on the condition
2. Used to Filter Rows
3. First table gets selected , then according to the where cluase condition some rows gets eleminated then selection of required columns is done
4. ***** make sure that “where” is not used Later “GroupBy” keyWord because for GroupBy we use Aggregate function like SUM/COUNT/AVERAGE etc .. for which we use “HAVING”
5. For where we use Conditional Statements
6. WHERE country=’Germany’ , score≠0, score>0

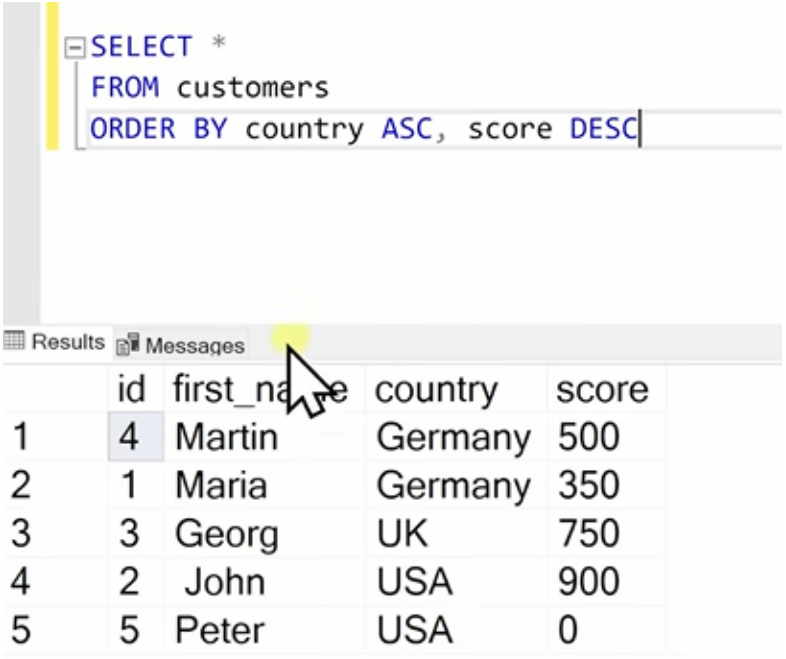
3. Order-By

1. This KeyWord is used to sort the data either in DESC/ASC
2. To filter the rows ,



4. Nested Order-By

1. Sorting of two columns at a time
2. Here the column name that written first have the more priority
3. Means ,first entered column will be sorted then, among the same column entity the score is sorted



	id	first_name	country	score
1	4	Martin	Germany	500
2	1	Maria	Germany	350
3	3	Georg	UK	750
4	2	John	USA	900
5	5	Peter	USA	0

a. Here see first country is written means first country column is sorted , then the score is sorted for those rows which have same country → sorted among the score..

```
SELECT *
FROM customers
ORDER BY score DESC, country ASC
```



esults Messages

id	first_name	country	score
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
1	Maria	Germany	350
5	Peter	USA	0

a. If we see here firs score is sorted then country is sorted among the score

b. here if we have same 900 score to two columns then country wise sorted will happens

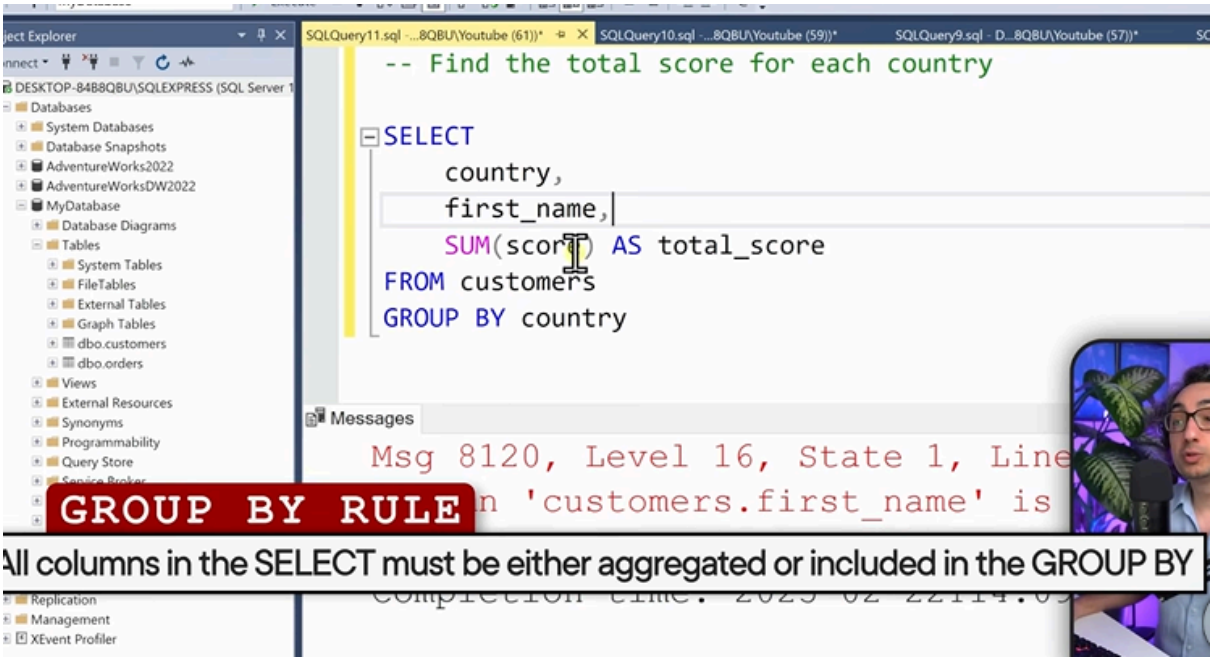
5. Group-By

1. Combines row with the same value (or)
2. Aggregates a column by another column
3. means if we have 3 rows with same country name , then then those 3rows will e combined to form a single row .. then what about if asked to retrieve country,score ...
4. Country gets aggregated then what about score→ we need only one value so conformly aggregate functions is needed → like sum(score) ⇒ all the rows those gets combined will finally have a column with score equals to sum all those combined rows kaa score



**** make sure the columns we are selecting while using the GROU_BY , must be a aggregate function kaa column or the grouped-by column only ****

**** because what if we used column aggregation and not used aggregation function to any column and asked to retrieve that column then what value will it give among thoses aggregated rows



1. error → because country→ india has 3 names→ a,b,c and combined using groupby ⇒ then finally we get india,(what name among a,b,c) ?? so error

```sql
-- Find the total score for each country

SELECT
    country,
    first_name,
    SUM(score) AS total_score
FROM customers
GROUP BY country,first_name
```

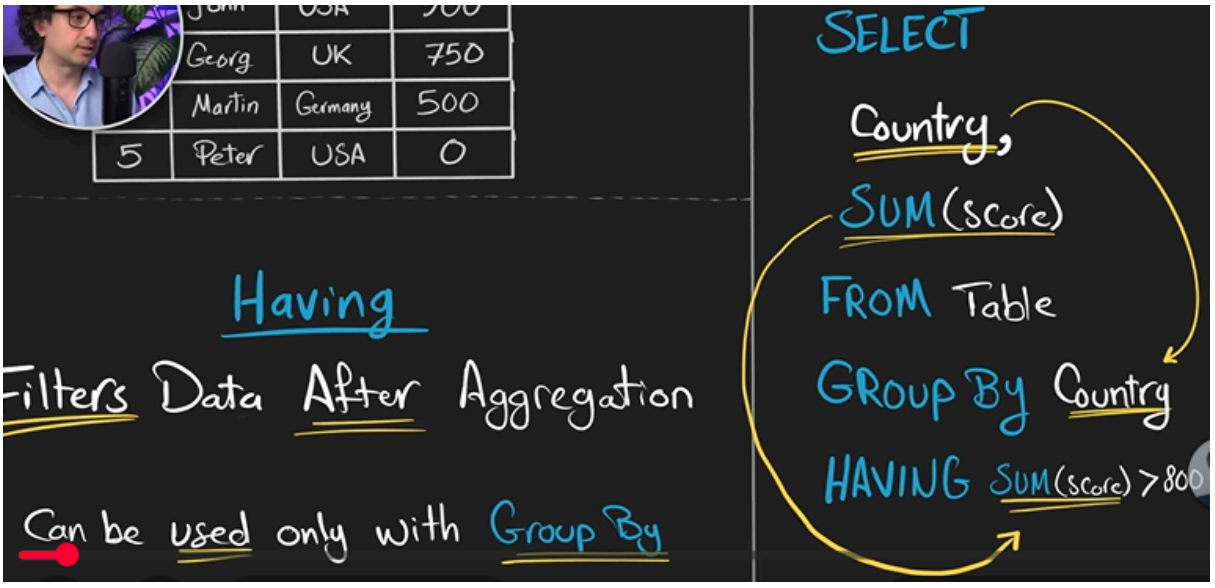
Results Messages

	country	first_name	total_score
1	USA	John	900
2	UK	Georg	750
3	Germany	Maria	350
4	Germany	Martin	500
5	USA	Peter	0

1. Here see we are aggregating based on country,first_name ⇒ means if rows having same country name and same first_name then only those gets aggregated

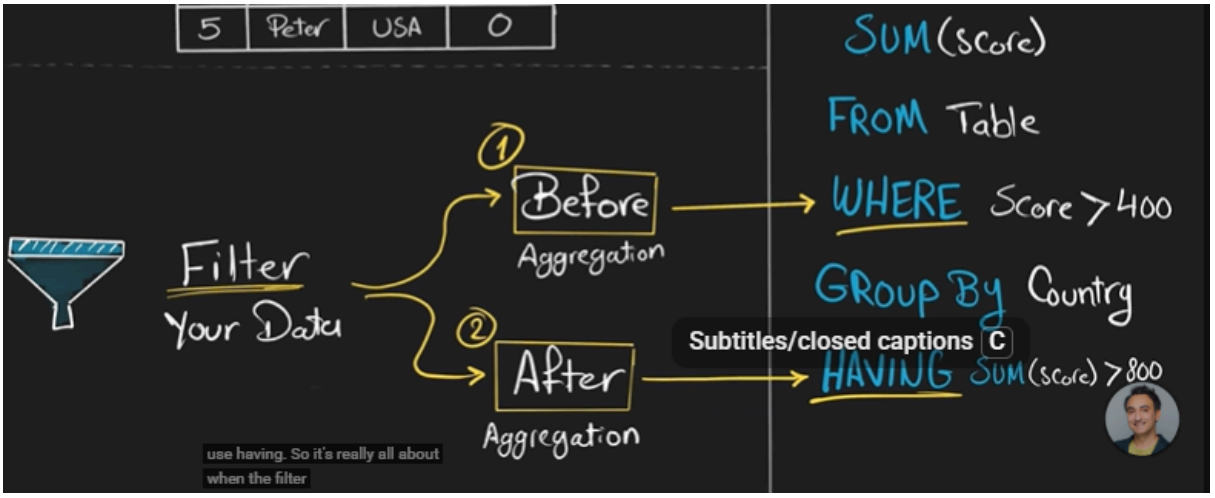
6. Having

1. used to Filter Aggregated Data ..
2. Having helps to filter rows after the use of Group-By keyword only



7. Where vs Having

1. First Where clause will be executed, so first accoding to Where condition the rows get filtered,then group-by → followed by Having



2. first table selected , then where → filters the rows , then group-by , then having , then selecting the columns

8. Distinct

1. Remove the Duplicates, retrieve a particular value only once from that column ⇒ same values will retrieved only once
2. DISTINCT slows down the query
3. The column name that we are writing in the Order-BY, must be in the distinct (if needed to use Distinct)

Why is this rule there?

After `DISTINCT country` is applied,

there is **no longer any single definite income value per country** —

because many people in the same country can have different incomes.

So SQL doesn't know:

> “Which person's income should I use to sort that country?”

4. Because after executing the DISTINCT keyword then orderBy is used

9. Top

1. This retrives the top values/rows of the table formed at last
2. this executed at last , (after remaining all executed and formed a table)
3.

```
SELECT TOP 3 *
FROM customers
ORDER BY score DESC
```

10. Static clause

1. to select any column with same name

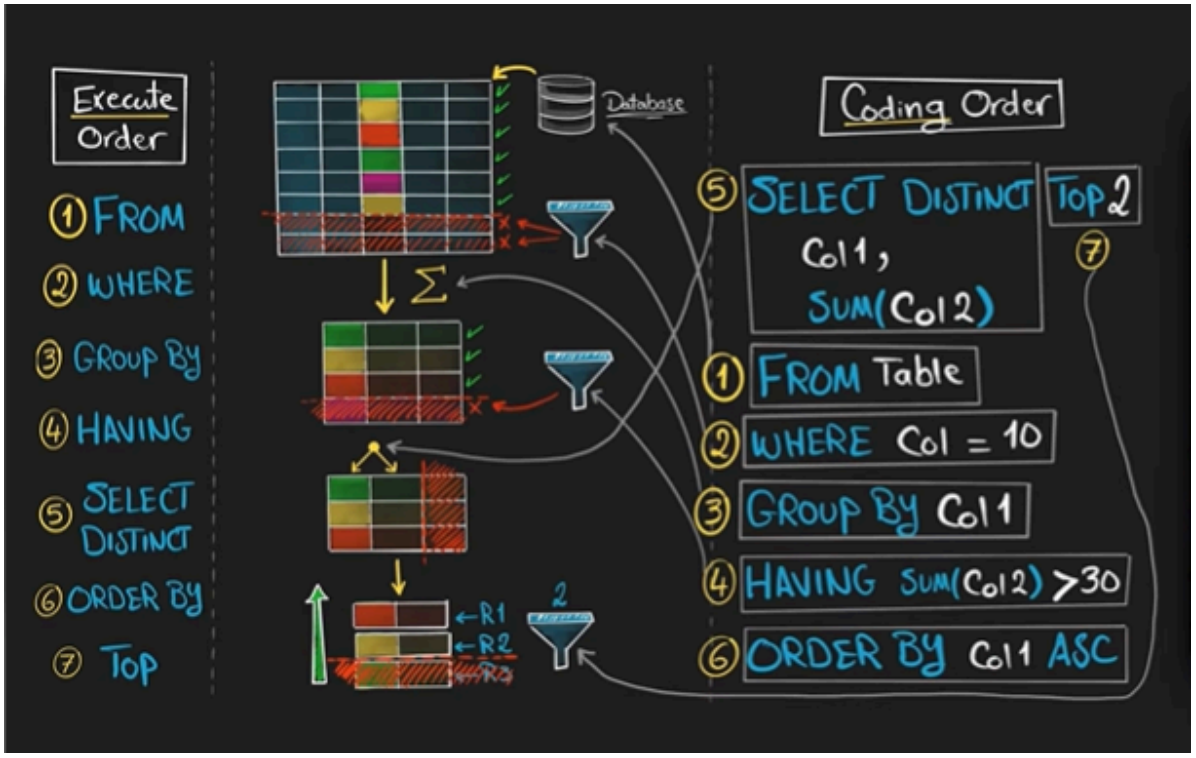
```
SELECT
id,
first_name,
'New Customer' AS customer_type
FROM customers
```

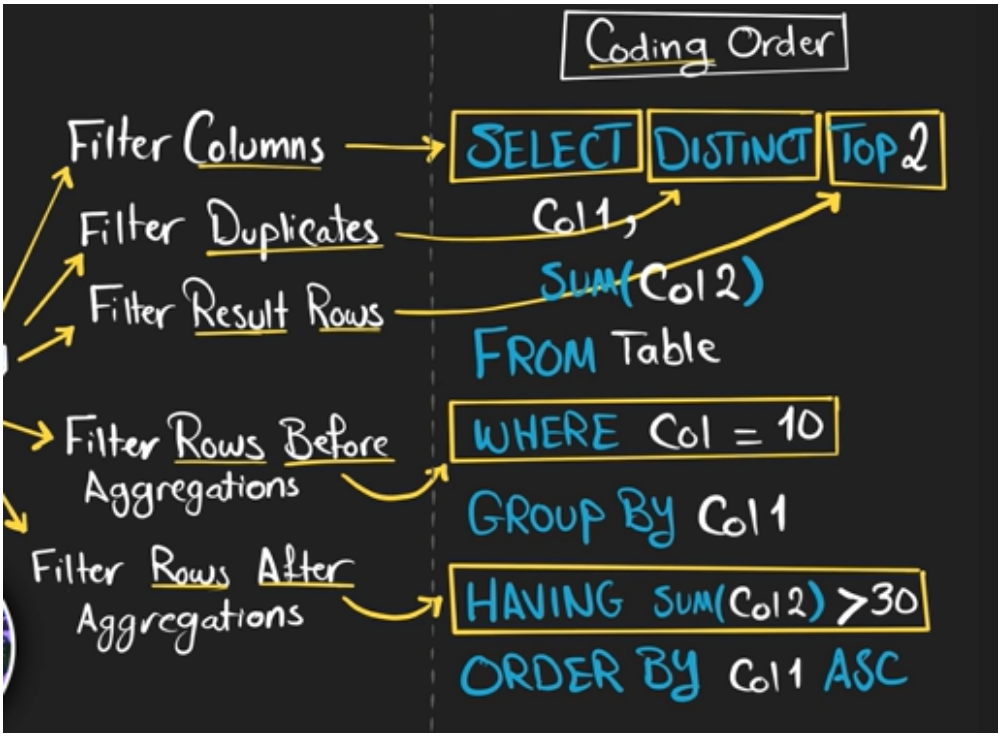
Messages

id	first_name	customer_type
1	Maria	New Customer
2	John	New Customer
3	Georg	New Customer
4	Martin	New Customer
5	Peter	New Customer



11. ** Order of Execution


Execute Order
1 FROM
2 WHERE
3 GROUP BY
4 HAVING
5 SELECT DISTINCT
6 ORDER BY
7 TOP
Database
Coding Order
5 SELECT DISTINCT Col1, SUM(Col2)
TOP 2
7
1 FROM Table
2 WHERE Col = 10
3 GROUP BY Col1
4 HAVING SUM(Col2) >30
6 ORDER BY Col1 ASC
R1
R2
2


Coding Order
Filter Columns
Filter Duplicates
Filter Result Rows
Filter Rows Before Aggregations
Filter Rows After Aggregations
SELECT DISTINCT TOP 2
Col1,
SUM(Col2)
FROM Table
WHERE Col = 10
GROUP BY Col1
HAVING SUM(Col2) >30
ORDER BY Col1 ASC